

Scalable and High-Fidelity Capture of Hand-Object Interactions

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Abstract. The growing demand for high-fidelity 4D hand-object interaction (HOI) data in embodied AI and spatial computing is currently bottlenecked by the reliance on pre-scanned object templates and physical markers. While recent methods have demonstrated promising results in reconstructing 4D hand-object interaction from videos, they are highly sensitive to initial estimates of hand and object poses. Yet, estimating these poses from images is challenging, in particular under severe occlusion which is inherent in hand-object interaction scenarios. We propose a novel system for the robust and accurate reconstruction of hands and objects from multi-view videos without requiring any templates or markers. Our system consists of two main components with key innovations: (1) a multi-view feed-forward transformer model that aggregates cross-view geometry and temporal cues to provide a reliable, metric-consistent initialization for both poses and dense object geometry, and (2) a hand-object physics-aware Gaussian-based optimization framework to refine the initial estimates, integrating tetrahedral constraints, collision refinement, and appearance decomposition to produce physically plausible and visually accurate reconstruction. Validated on public benchmarks and an extensive internal dataset, our pipeline achieves highly robust, artifact-free reconstruction, providing a scalable foundation for generating physically consistent digital assets.

Keywords: 4D Reconstruction · Hand-Object Interaction · Gaussian Splatting · Spatiotemporal Prior · Digital Assets

1 Introduction

The rapid evolution of embodied AI and immersive spatial computing has catalyzed a surging demand for high-fidelity 4D hand-object interaction (HOI) data [3, 10, 15–17, 19, 20, 45, 58]. Conventionally, capturing such data requires known object templates and physical markers [15] or manually annotated key-points [3] to track the object and hand poses. This strict dependency prevents these methods from generalizing to diverse, unknown objects in the wild, severely bottlenecking the scalability of existing capture pipelines.

Several recent methods have attempted to reconstruct 4D interacting hands and objects from video alone [8, 14, 16, 38], bypassing markers and templates by optimizing 3D Gaussian or NeRF representations. While promising, these methods are highly sensitive to initialization. An inaccurate estimate of initial hand

or object poses often leads the optimization toward poor local minima or total divergence. Despite rapid progress [7, 46, 53, 61], estimating hand and/or object poses from images remains challenging due to inherent depth ambiguity and severe dynamic occlusions. Consequently, existing video-based capture methods often lack robustness or must impose restrictive assumptions, such as requiring objects to remain static [38] or move at low velocities [8].

To bridge this gap, we propose a novel system capable of reconstructing hands and unknown objects during natural interactions from multi-view video with significantly improved robustness and accuracy. Our framework adopts a two-stage design with key innovations at each phase. We first employ a novel multi-view, spatiotemporal feed-forward model for robust initialization, followed by a physics-aware Gaussian-based optimization framework that refines geometry and appearance.

Our first innovation addresses the initialization bottleneck via the hand object spatiotemporal transformer (HOST). Existing hand or object pose estimators [7, 46, 53, 61] are primarily designed for monocular input and rely heavily on learned priors, which limits their reliability in complex interaction scenarios. While recent foundational 3D models (e.g., VGGT [49]) effectively leverage multi-view cues, they are generally tailored for reconstruction of static scenes. HOST extends these foundational models to the task of multi-view tracking of dynamic interacting hands and objects. By introducing a factorized spatiotemporal attention mechanism, HOST efficiently aggregates information across viewpoints and frames, producing accurate and temporally coherent object and hand pose estimates. In addition, even if the object is partially occluded in some frames, HOST is able to reconstruct a complete and dense point cloud by using the geometric cues in all frames and viewpoints. HOST effectively resolves depth ambiguities where monocular trackers fail, ensuring metric 3D consistency and providing a reliable geometric foundation for subsequent refinement.

Our second innovation is a hand-object physics-aware Gaussian (HOPG) optimization framework, which elevates this initialization into high-fidelity 4D assets. We model the two hands and the unknown object with independent 2D Gaussian [24] clouds, which are initialized with the tracked pose parameters as well as the reconstructed dense point cloud from HOST. We optimize the 2D Gaussians along with the object and hand poses by minimizing the photometric loss. To prevent implausible distortions or volume collapse of the hand during close interactions [8, 37], we anchor the hand Gaussian on a tetrahedral mesh proxy and regularize the optimization with a volumetric As-Rigid-As-Possible (ARAP) energy [44]. Furthermore, we incorporate a dedicated physics-aware collision resolution step to ensure no penetration. To reconstruct high-fidelity appearance of hand and object, we further introduce a temporal batching strategy and a decoupled model [42] that disentangles view-independent albedo and view-dependent specular. This decomposition prevents environmental lighting from being “baked” into the geometry, recovering the intrinsic surface colors.

Leveraging our framework, we processed a large-scale dataset of 77 complex sequences involving 32 unique objects with complex geometries. Achieving

an automated success rate exceeding 93%, we demonstrate our system’s exceptional robustness and scalability for automated 4D asset generation. We extensively evaluate both our isolated modules (HOST and HOPG) and the full pipeline across a public benchmark [34], the VEPHand dataset [1], and our custom dataset. The results demonstrate the effectiveness of our framework across three core tasks: it establishes superior robust pose tracking, excels in dynamic novel view synthesis, and recovers highly accurate dense geometry.

In summary, our key contributions are:

- A system that robustly and accurately reconstructs hand object interaction from multi-view videos without requiring pre-scanned templates or known object and hand poses, enabling scalable hand object interaction data capture.
- A spatiotemporal transformer model that robustly infers interacting poses and shapes from multi-view inputs, providing a reliable geometric initialization even under severe occlusion.
- A physics-aware Gaussian optimization framework integrating tetrahedral constraints, collision refinement, and appearance decomposition to produce physically plausible and visually accurate 4D reconstructions.

2 Related Works

Neural Hand and HOI Rendering. Reconstructing high-fidelity dynamic hand-object interactions (HOI) is a long-standing challenge. To address this, foundational work first focused on bare-hand neural rendering. Early neural implicit methods, such as HandAvatar [5], achieved high fidelity by disentangling hand geometry and texture but suffered from slow rendering speeds. While subsequent works like LiveHand [35] overcame this to achieve real-time performance via mesh-guided sampling, 3D Gaussian Splatting (3DGS) has recently emerged as a fundamentally more efficient alternative. GauHuman [23] models articulated humans by warping 3D Gaussians via Linear Blend Skinning, and HandSCS [12] explicitly handles severe hand articulations by introducing a Structural Coordinate Space to bridge sparse skeletal joints and dense Gaussians.

Building upon these bare-hand reconstruction foundations [12, 18, 26], a substantial body of work [56, 60] tackles HOI through per-scene optimization to recover precise surfaces and contact mechanics. Recent methods achieve this using articulated 3D Gaussians (MANUS [38], DyTact [8]), compositional implicit fields (HOLD [14]), or generative diffusion priors for unobserved regions [33, 50, 64]. However, these frameworks suffer from two critical limitations: they inevitably collapse when their required tracking initializations drift under severe occlusions, and they frequently “bake” environmental illumination into the geometry.

Hand-Object Pose Tracking. To provide the tracking priors required by the aforementioned optimization pipelines, an emerging line of inference-based methods trains feed-forward models to directly regress hand and object states from

images. For instance, HORT [7] employs a coarse-to-fine transformer architecture to efficiently reconstruct dense 3D point clouds of hand-held objects and their hand-relative poses. For complex hand motions, Dyn-HaMR [61] recovers global 4D trajectories from monocular videos by combining hierarchical hand tracking with generative motion priors and SLAM. Concurrently, foundation models like SAM 3D [4] have generalized single-image 3D perception, demonstrating remarkable capabilities in predicting object geometry, texture, and scene layout directly from in-the-wild 2D images.

However, their reliance on monocular inputs makes them highly susceptible to occlusions and depth ambiguities, rendering them too fragile to initialize complex 4D reconstructions. In contrast, our HOST module aggregates cross-view and temporal cues to resolve these depth ambiguities, establishing a metrically reliable initialization for the downstream optimization.

Feed-Forward Geometric Estimation. The success of large-scale pre-training has shifted 3D geometric estimation toward feed-forward regression [22, 52]. Pioneered by DUST3R [31, 51, 57], foundational works demonstrate that dense 3D point maps can be directly regressed from unposed images. Building upon this, VGGT [49] and its successors [43, 55] introduced large networks [13] for joint camera and scene geometry prediction. To adapt these spatial priors for dynamic 4D environments without incurring prohibitive computational overhead, recent works like 4D-VGGT [48] and MonST3R [62] introduce spatiotemporal decoupling to model structural consistency and motion continuity separately. Unlike these foundational models, which primarily output generic and unconstrained scene point clouds, HOST is explicitly designed to parse complex interactions. By introducing a spatiotemporal attention mechanism, we efficiently aggregate multi-view and temporal cues to jointly regress hand and object states and entity-specific dense geometry. This effectively translates powerful generic 3D priors into a specialized articulated tracker, providing a reliable 4D initialization for our downstream HOPG pipeline.

3 Method

Given synchronized multi-view RGB videos and calibrated camera parameters, our framework first employs the Hand Object Spatiotemporal Transformer (Sec. 3.1) to establish a robust multi-view spatiotemporal prior, followed by a Hand Object Physics-aware Gaussian (HOPG) refiner (Sec. 3.2). The full pipeline is illustrated in Fig. 1.

3.1 HOST: Hand-Object Spatiotemporal Transformer

The first stage of our framework establishes a robust geometric prior to guide the downstream volumetric refinement. Given a sequence of F frames across V

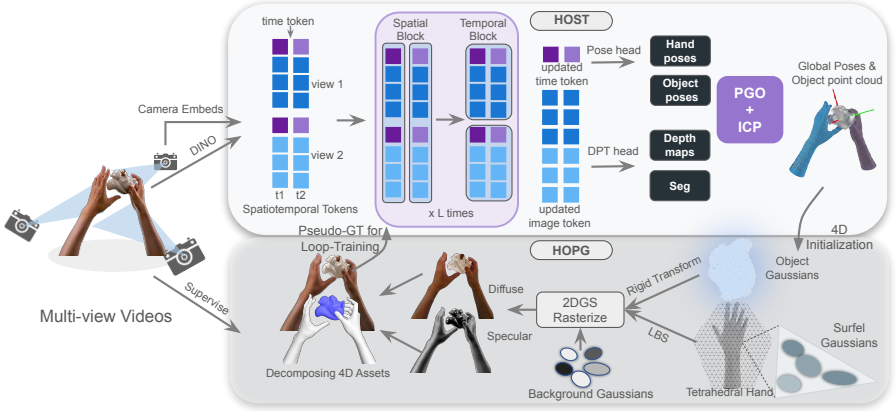


Fig. 1: Overview of our framework. Our pipeline operates in two stages. First, the Hand Object Spatiotemporal Transformer (HOST) processes multi-view videos to robustly regress parametric hand/object poses, dense object point clouds, and segmentation masks, yielding a metric 3D initialization. Second, the Hand Object Physics-aware Gaussian (HOPG) module leverages this initialization to optimize a hybrid 2D Gaussian representation. By enforcing structural constraints and explicitly decoupling diffuse and specular appearance, HOPG effectively prevents illumination bake-in, reconstructing high-fidelity 4D assets.

synchronized static views, HOST regresses temporally coherent 4D attributes and dense geometric features. Formally, HOST defines a mapping $\mathcal{F}$ as follows:

$$(\mathbf{D}_{1:F,1:V}, \boldsymbol{\Theta}_{1:F}^h, \boldsymbol{\Theta}_{1:F}^o) = \mathcal{F}(\mathbf{I}_{1:F,1:V}, t_{1:F}, \mathbf{E}_{1:V}, \mathbf{K}_{1:V}), \quad (1)$$

where $\mathbf{D}_{1:F,1:V}$ denote the dense predictions (i.e., semantic segmentation masks, depth maps, and point maps) across all F frames and V viewpoints; $\boldsymbol{\Theta}_{1:F}^h$ and $\boldsymbol{\Theta}_{1:F}^o$ refer to the full sequences of hand and object parameters, respectively; $\mathbf{I}_{1:F,1:V}$ denotes the input multi-view video sequence; $\mathbf{E}_{1:V}$ and $\mathbf{K}_{1:V}$ denote the camera extrinsic and intrinsic parameters for all views, and $t_{1:F}$ represent the relative frame identifiers.

Spatiotemporal Feature Backbone. HOST builds upon the VGGT [49] transformer backbone but extends it to a spatial and temporal structure. To encode spatial and temporal information, each image $\mathbf{I}_{f,v}$ is patchified into tokens $\mathbf{T}_{\text{image}}$ using a frozen DINOv2 [36] backbone, and augmented with view-consistent, ray-based camera embeddings [27]. Simultaneously, relative frame identifiers t_f are mapped to view-shared time tokens $\mathbf{T}_{\text{time}}$, which are then concatenated with $\mathbf{T}_{\text{image}}$ to form the final input sequence $(\mathbf{T}_{\text{time}}, \mathbf{T}_{\text{image}})$ for the factorized attention layers.

Processing massive 4D sequences (V views, F frames, N patches) via monolithic global attention, such as the frame-global loop in VGGT, incurs a prohibitive computational complexity of $\mathcal{O}((VFN)^2)$. To circumvent this bottleneck while preserving rich 4D context, we propose the spatiotemporal mechanism. By retaining foundational intra-frame attention but explicitly factorizing the in-

tractable global attention into alternating Spatial and Temporal Blocks (Fig. 1), the feature update cycle for the k -th group is formulated as:

$$\mathcal{B}_{\text{spatial}} : \mathbf{T}'_k = \mathcal{A}_{\text{view}}(\mathcal{A}_{\text{frame}}(\mathbf{T}_k)), \quad (2)$$

$$\mathcal{B}_{\text{temporal}} : \mathbf{T}_{k+1} = \mathcal{A}_{\text{temporal}}(\mathcal{A}_{\text{frame}}(\mathbf{T}'_k)), \quad (3)$$

Here, $\mathcal{A}_{\text{frame}}$ performs local intra-frame self-attention (costing $\mathcal{O}(VFN^2)$). The Spatial Block ($\mathcal{B}_{\text{spatial}}$) aggregates stereo cues via cross-view attention $\mathcal{A}_{\text{view}}$ for each frame independently, bounding its cost to $\mathcal{O}(F(VN)^2)$. Conversely, the Temporal Block ($\mathcal{B}_{\text{temporal}}$) aggregates motion cues via temporal attention $\mathcal{A}_{\text{temp}}$ for each view independently, costing $\mathcal{O}(V(FN)^2)$. This factorized design successfully aggregates global 4D context while drastically reducing the quadratic computational overhead. In practice, we train the framework using a chunking strategy, e.g., a typical chunk we use in training is 5 frames with 6 views.

Dense Geometry Heads. We employ a Dense Prediction Transformer (DPT) architecture [39] to decode the features into a set of per frame and per view dense maps $\mathbf{D}$: multi-class probability map for the left hand, right hand, and the interacting object; depth map and point map. This joint supervision encourages the network to disentangle the tightly interacting entities implicitly, providing a clean, entity-specific geometric foundation.

Global Kinematics Heads. To estimate the global 4D kinematic state, we pass the final stage time tokens $\hat{\mathbf{T}}_{\text{time}}$, which encode the overall spatiotemporal dynamics, to dedicated MLP heads that regress the per-frame kinematic states for hand and object parameters. The hand parameters $\Theta^h = \{\alpha, \beta\}$ here refer to the identity α and poses β w.r.t. a given parametric hand model [41].

For the rigid object transform, we adopt a hybrid state representation comprising the absolute centroid translation $\mathbf{t}_f^{\text{abs}}$ in world coordinates and the relative rotation $\Delta \mathbf{R}_{1 \rightarrow f}$ (with respect to the first frame of the input chunk). This asymmetric design explicitly addresses the distinct observability of translation and rotation. Because the canonical orientation of an unseen object is inherently ambiguous without a predefined template, predicting relative rotation is strictly necessary. While it may seem intuitive to also predict relative translations, integrating frame-to-frame offsets inevitably accumulates trajectory drift over time. Instead, since the object’s absolute 3D geometric center remains continuously observable from spatiotemporal cues, we directly regress its absolute translation. This firmly anchors the spatial trajectory, eliminating long-term positional drift and confining any unavoidable error accumulation strictly to the rotational domain.

Training Losses. We follow the VGGT [49] for dense prediction (segmentation, depth and point map) training and apply geodesic distance [21, 25] on rotation and L2 losses on translation and registered meshes from forward kinematics.

Inference. To process arbitrarily long videos under memory constraints, we split sequences into overlapping chunks using a sliding window with a stride of one. For hands, simply averaging the overlapping predictions provides sufficient

temporal smoothing. For rigid objects, however, we prevent long-term trajectory drift by aggregating chunk-wise predictions in two steps:

We first employ *Pose Graph Optimization (PGO)* [11, 30] to combine per-window pose estimates into a continuous trajectory. Specifically, we optimize absolute object pose estimate $\Theta^o = (\mathbf{R}, \mathbf{t})$ with the predicted relative rotation edges $\mathcal{E}$ and absolute translations:

$$\arg \min_{\{\mathbf{R}_f, \mathbf{t}_f\}} \sum_{(i,j) \in \mathcal{E}} \|\mathbf{R}_i^T \mathbf{R}_j - \Delta \mathbf{R}_{i \rightarrow j}\|_F^2 + \lambda_{\text{abs}} \sum_f \|\mathbf{t}_f - \mathbf{t}_f^{\text{abs}}\|^2. \quad (4)$$

To eliminate residual drift, we then unproject the masked depth maps into 3D point clouds and align them via frame-to-model ICP [2, 65] with the PGO trajectory as a robust initialization. In this way, we are able to infer accurate object pose trajectories even in long videos. Fusing these aligned observations yields the canonical object point cloud $\mathbf{V}_{\text{obj}}^{\text{cano}}$.

In summary, this inference strategy aggregates per-window predictions into globally consistent hand parameters Θ^h , absolute object poses Θ^o , and an optional canonical point cloud of the object $\mathbf{V}_{\text{obj}}^{\text{cano}}$. These outputs are then used to initialize the subsequent HOPG refinement.

3.2 HOPG: Hand-Object Physics-aware Gaussian Optimization

In this stage, we jointly optimize three distinct 2D Gaussian fields, hand, object, and background, leveraging the pose and shape estimates from HOST as an initialization.

Volumetric Hand. We utilize a volumetric deformation model inspired by VEPHand [1] as our geometric proxy. To faithfully model volume preservation and muscle bulging and to resolve collision during close interactions, we embed the hand surface within a canonical tetrahedral mesh. However, rather than optimizing per-frame volumetric variables, we predict these intrinsic tetrahedral deformations dynamically via a small MLP network.

Specifically, let $\mathbf{v}_c$ denote the canonical surface vertices and $\boldsymbol{\rho}_c$ represent the vertices of the underlying canonical tetrahedral mesh. Unlike standard parametric models that apply Linear Blend Skinning (LBS) directly to the base surface, our volumetric model first applies non-rigid, pose-dependent deformations in the tetrahedral domain:

$$\text{(Standard Surface)} \quad \mathbf{v}_p = \text{LBS}(\mathbf{v}_c, \alpha, \beta), \quad (5)$$

$$\text{(Our Volumetric)} \quad \boldsymbol{\rho}'_c = \boldsymbol{\rho}_c + \text{MLP}(\boldsymbol{\rho}_c, \beta), \quad (6)$$

$$\mathbf{v}'_c = \mathcal{M}_{\text{tet} \rightarrow \text{surf}}(\boldsymbol{\rho}'_c), \quad (7)$$

$$\mathbf{v}_p = \text{LBS}(\mathbf{v}'_c, \alpha, \beta), \quad (8)$$

where α and β represent the identity shape and kinematic pose parameters, respectively. In our formulation, a lightweight MLP predicts pose-dependent offsets for the tetrahedral control points to yield the deformed volume $\boldsymbol{\rho}'_c$. We then compute the updated canonical surface vertices $\mathbf{v}'_c$ via a predefined barycentric

mapping operator $\mathcal{M}_{\text{tet} \rightarrow \text{surf}}$. Geometrically, each surface vertex is initialized inside a specific tetrahedron. This operator calculates the deformed surface position simply by interpolating the displaced tetrahedral vertices ρ'_c using fixed barycentric weights. Consequently, the surface acts as a continuous skin strictly bound to the underlying volumetric flesh. Finally, standard Linear Blend Skinning (LBS) articulates these volume-aware canonical surface vertices into the final posed space $\mathbf{v}_p$.

Following PICA [37], we anchor 2D Gaussians to this deformed mesh using barycentric interpolation, ensuring the radiance field strictly inherits the kinematic consistency of the hand model.

Rigid Object. The object is represented by a set of canonical 2D Gaussians, seamlessly initialized from the fused canonical point cloud $\mathbf{V}_{\text{obj}}^{\text{cano}}$ provided by HOST. During refinement, we jointly optimize these canonical Gaussian attributes and the per-frame rigid transformations.

Background. We model the static environment using a separate set of 2D Gaussians initialized on a large enclosing sphere.

Appearance Model. Standard Gaussian Splatting [28] bakes view-dependent lighting into the spherical harmonics (SH) of each primitive. In highly dynamic hand-object interactions, this entanglement between extreme rotations and illumination causes severe flickering and texture baking. To reconstruct physically and temporally consistent assets, we explicitly factorize intrinsic albedo, diffuse shading, and specularity. Following Saito et al. [42], the dynamic diffuse color $\mathbf{C}_{\text{diff}}$ of a Gaussian is formulated by convolving environmental light SH coefficients $\mathbf{L}$ with dynamically rotated canonical transfer coefficients $\mathbf{d}(\mathbf{R})$, modulated by a learnable view-independent albedo $\mathbf{a}$:

$$\mathbf{C}_{\text{diff}} = \mathbf{a} \odot \sum_{i=1}^{(n+1)^2} \mathbf{L}_i \mathbf{d}_i(\mathbf{R}), \quad (9)$$

where n is the SH degree, and $\mathbf{R}$ defines the Gaussian’s rotation from canonical to world space. Explicitly applying $\mathbf{R}$ ensures the transfer coefficients remain aligned with the global environment light, effectively decoupling illumination from the primitive’s motion. Furthermore, to prevent view-dependent highlights from polluting this intrinsic albedo, we model specularity using a dedicated lightweight MLP [47]. The network processes the Integrated Directional Encoding (IDE) of the reflection vector $\mathbf{r}$ and the Gaussian’s learnable surface roughness ρ to output a specular residual. The final composited color is then evaluated as $\mathbf{C}_{\text{total}} = \mathbf{C}_{\text{diff}} + s \cdot \text{MLP}_{\text{spec}}(\text{IDE}(\mathbf{r}, \rho))$, where s is a learnable, primitive-specific specular intensity.

Losses. The data terms are driven by standard photometric losses (L1, D-SSIM, LPIPS). For physical fidelity, we impose an As-Rigid-As-Possible (ARAP) energy [44] on the underlying tetrahedral mesh for volume preservation on deformed canonical tetrahedra ρ'_c and a ray-tracing based collision loss [1] on posed surface mesh $\mathbf{v}_p$ to prevent both inter and self penetration.

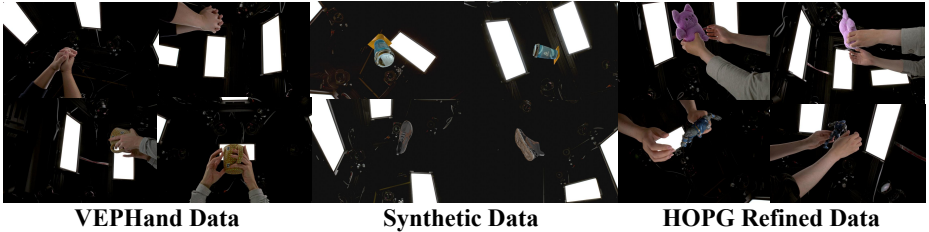


Fig. 2: Overview of our composite training dataset. Left: Multi-view captures from the VEPHand [1] dataset. **Middle:** Synthetic data rendered in Blender using diverse textured objects from Texverse [63] Dataset. **Right:** Custom captures refined via our HOPG module.

3.3 Dataset Roadmap

To ensure kinematic diversity and geometric generalizability, HOST is trained on a comprehensive composite dataset. The foundation comprises 1,600 multi-view sequences from VEPHand Dataset [1], featuring high-quality scans and hand-object registrations across 17 3D-printed CAD models. To overcome the limited object diversity of this set, we synthesize a supplementary dataset in Blender, rendering 500 high-resolution textured objects [63] simulated with randomized 6D trajectories. Finally, to capture real-world complexity, we collect a custom dataset of 47 everyday objects augmented with miniature ArUco markers to reliably acquire initial relative poses. These captures are then processed by our HOPG refiner. Once converged, HOPG acts as a robust pseudo-label generator, yielding refined ground-truth equivalents (cleaner masks, accurate depth, and collision-free poses). These high-fidelity annotations are then fed back to fine-tune HOST, progressively closing the domain gap without manual intervention.

4 Experiment

To comprehensively evaluate our framework’s capabilities in photorealistic novel view synthesis (NVS), physically plausible geometry recovery, and industrial-level scalability, we conduct extensive experiments on a composite evaluation suite comprising both public benchmarks and diverse in-house captures.

4.1 Experimental Setup

Implementation Details HOST is built on a ViT-Large backbone with frozen DINOv2 [36] patch embeddings and initialized with pre-trained VGGT [49] weights. We optimize the network using AdamW in bfloat16 precision. To stabilize early training, we introduce a 2,000-step geometry warmup where we only optimize the backbone and dense geometry heads. The model is first pre-trained for 200k steps on 64 TPU v5 accelerators ($lr = 8.0 \times 10^{-5}$) on VEPHand data, and subsequently fine-tuned for 20k steps on 8 NVIDIA H100 GPUs ($lr = 1.0 \times 10^{-5}$) on synthetic and our HOPG refined data. For the HOPG refinement, we use a batch size of 5 frames to explicitly enforce the decoupled appearance model. A

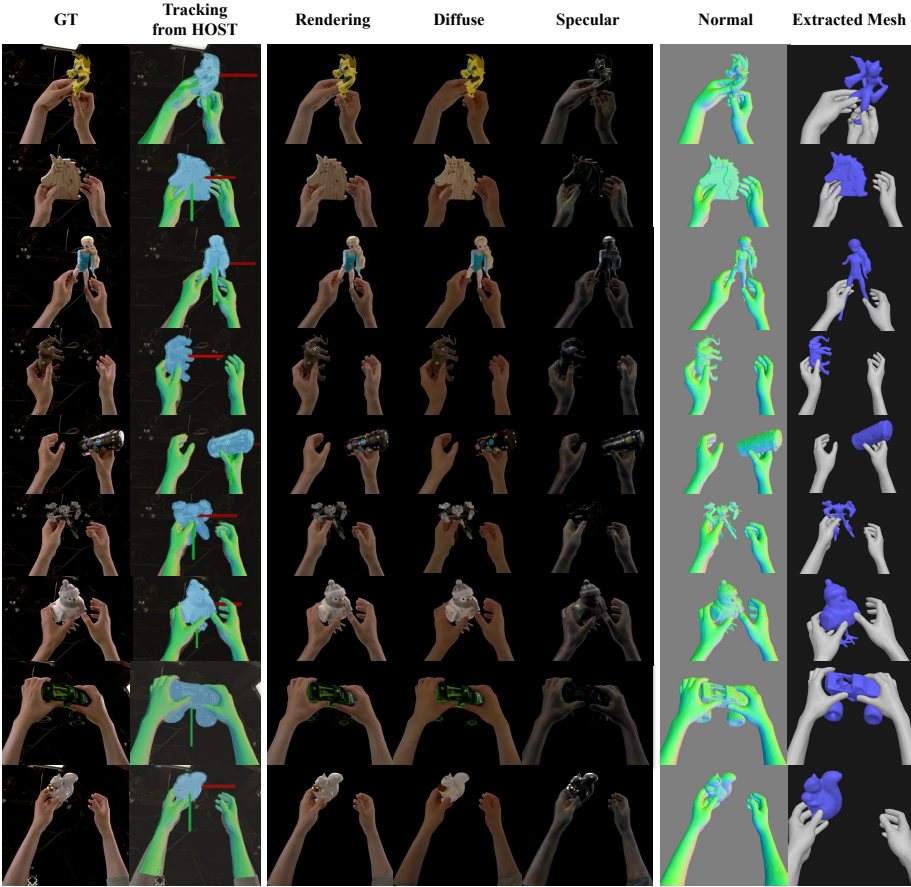


Fig. 3: Automated, Scalable High-Fidelity Capture across Diverse Objects. Given calibrated multi-view images (GT), HOST provides robust tracking initializations. HOPG then achieves photorealistic rendering by decomposing appearance into diffuse and specular components. This explicit lighting decoupling prevents artifacts from baking into the geometry, yielding clean normal maps and high-fidelity 3D meshes.

typical sequence of 21 views and 600 frames takes about 30k iterations (roughly 5 hours) to converge on a single NVIDIA A100 GPU.

Scalability Validation. To validate scalability, we deploy our pipeline on a newly captured multi-view dataset of diverse objects. As qualitatively demonstrated in Fig. 3, our framework operates entirely automatically, successfully reconstructing 72 out of 77 interaction sequences across 32 unique objects, thereby achieving an exceptional $\sim 93.5\%$ success rate. Furthermore, we also showcase the high-resolution 3D object meshes extracted via TSDF fusion [9] in Fig. 4. The few failures stem from challenging object properties, such as geometrically thin structures (pens) and highly symmetric, textureless surfaces (tapes), which induce severe pose ambiguities during HOST initialization.

Evaluation Datasets. We comprehensively evaluate our framework across three distinct domains. First, a representative 10-sequence subset from our custom cap-

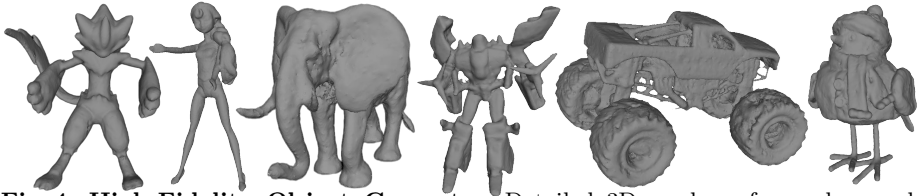


Fig. 4: High-Fidelity Object Geometry. Detailed 3D meshes of complex, real-world objects extracted via TSDF fusion.

tures serves as the primary testbed for overall pipeline comparisons. Second, we benchmark our isolated HOPG module on InterHand2.6M [34]. Finally, to rigorously evaluate both our feed-forward initialization and the final dense geometry reconstruction, we utilize the VEPHand Dataset [1], which provides high-quality per-frame volumetric pseudo-ground truth across 7 complex sequences.

4.2 Novel View Synthesis

In this section, we first evaluate novel view synthesis capability of HOPG on InterHand2.6M and then assess the full pipeline on our Custom Dataset.

HOPG on InterHand2.6M. We follow the HandSCS [12] protocol (10 training, 15 held-out views) across their exact 9 selected sequences. As shown in Table 1 (left), our method is highly competitive. Standard NeRf [6,35] or 3DGS baselines [12] artificially inflate photometric scores by relying on semi-transparent volume accumulation and baked-in illumination. In contrast, our formulation trades unconstrained photometric memorization for physical accuracy by enforcing strict, opaque 2D surfaces and explicitly disentangling albedo from specular components. We also provide qualitative rendering results in Fig. 6a.

Full pipeline on Custom Data. To evaluate the complete framework, taking only raw multi-view videos and camera parameters as input, we compare against representative dynamic scene methods: 4DGS [54] and Deformable 3DGS [59]. To focus strictly on the interaction, we utilize SAM2 [29,40] to isolate the foreground hand and object. As reported in Table 1 (right), our explicitly constrained representation outperforms these baselines by a massive margin (> 4.5 dB in PSNR). Qualitatively (Fig. 5), general dynamic methods lack structural priors, causing their optimization to degrade into severe motion blur and catastrophic geometric collapse under highly non-linear kinematics. Conversely, guided by the accurate metric priors from HOST, our framework seamlessly tracks these motions while maintaining sharp visual boundaries.

4.3 Geometry Evaluation

To verify that our framework recovers 3D geometry rather than merely overfitting photometric projections, we evaluate it on the VEPHand Dataset with per-frame GT scan. We report the one-way Surface Accuracy (point-to-surface distance from the predicted vertices to the ground-truth scans, which include

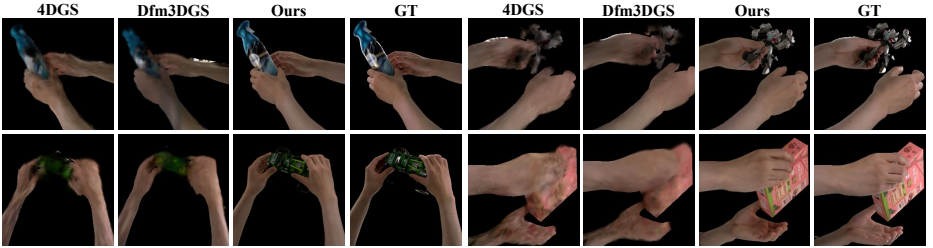


Fig. 5: Qualitative comparison on our Custom-Captured Dataset. Our method successfully tracks difficult motions and recovers complex object geometries, whereas prior-free baselines suffer from severe motion blur and geometric collapse.

Table 1: Quantitative NVS results. Comparison against specialized hand avatars on InterHand2.6M (left, 10-view train / 15-view test) and general dynamic baselines on our Custom-Captured Dataset (right, 17-view train / 4-view test).

InterHand2.6M				Custom Dataset			
Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LiveHand [35]	31.06	0.977	0.0384	4DGS	26.34	0.943	0.0814
HandAvatar [6]	32.14	0.962	0.0437	Dfm3DGS	26.32	0.947	0.0809
GauHuman [23]	31.73	0.957	0.0462	Ours (Full)	30.86	0.968	0.0370
HandSCS	35.51	0.974	0.0292				
Ours (HOPG)	35.07	0.970	0.0348				

the forearm). Specifically, we compare our HOST initialization against state-of-the-art feed-forward trackers, and assess the final dense geometry refined by our full pipeline (Table 2).

We first evaluate feedforward approaches by comparing our HOST results against state-of-the-art pose estimators under a sparse 6-view setup. We compare against HORT [7] and a composite baseline comprising DynHaMR [61] for hands and MV-SAM3D [32] for objects. To establish an optimally competitive multi-view baseline for these methods, we provide them with significant informational advantages. For the inherently monocular hand trackers (HORT and DynHaMR), we elevate their predictions into metric space using ground-truth camera parameters, aggregating their root-relative articulations via a robust 3D spatial median across all 6 views followed by a rigid registration step. For the object reconstruction baseline, we utilize MV-SAM3D, an open-source multi-view extension. To ensure a highly competitive evaluation, we explicitly provide it with ground-truth camera parameters and depth maps. This allows the model to reliably determine cross-view visibility, enabling it to effectively handle occlusions.

Despite arming these baselines with optimal 3D lifting and pseudo-ground-truth depth, our HOST module outperforms them end-to-end (achieving 3.07 mm vs. 5.81 mm for the composite baseline, and 7.22 mm for HORT). This demonstrates that our factorized spatiotemporal attention effectively leverages continuous 4D context to maintain accurate and stable trajectories, whereas isolated multi-view trackers can still struggle with drift. Building upon this robust

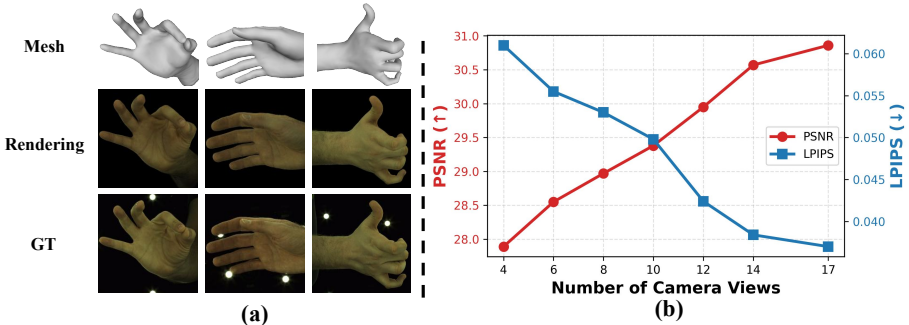


Fig. 6: InterHand2.6M Validation and View Robustness. (a) Our HOPG module achieves high-fidelity rendering and accurate geometry on bare-hand benchmarks. (b) End-to-end reconstruction remains highly stable across varying numbers of input cameras, maintaining strong accuracy even under extreme sparsity (4 views).

Table 2: Geometry Evaluation. One-way Surface Accuracy (mm, ↓) evaluated on the VEPHand Dataset. Our HOST module provides a superior metric initialization compared to feed-forward baselines, while our full pipeline further refines the geometry to achieve highly competitive dense accuracy. Processing time denotes the average time per frame measured on a single NVIDIA A100 GPU.

Method	Accuracy (mm) ↓	Processing Time ↓
HORT [7]	7.22	2s
DynHaMR [61] + SAM3D [32]	5.81	1min
HOST (Ours)	3.07	2s
Full Pipeline (Ours)	2.35	0.5min

initialization, our HOPG module successfully refines the dense geometry, further reducing the surface error to 2.35 mm.

4.4 Ablation Studies

We investigate the impact of our core design components on our Custom-Captured Dataset. A comprehensive analysis of our physical constraints and collision optimization is presented in Fig. 7.

Physics-Aware Geometry Constraint. We evaluate the necessity of our tetrahedral representation (Tet + 3D ARAP) compared to a standard surface-based approach (Surf + 2D ARAP). Since our pipeline operates entirely without ground-truth segmentation masks, optimizing exclusively via photometric rendering loss exposes the critical weakness of 2D ARAP: its inability to constrain internal volume, as illustrated in Fig. 7(a). In contrast, computing the ARAP energy over the 3D tetrahedral domain explicitly locks the internal structure. This rigorously preserves anatomical thickness, successfully preventing both texture-driven flattening and background-induced geometric collapse.

Collision Optimization. While photometric losses ensure visual alignment, they cannot constrain invisible regions. As shown in Fig. 7(b), our offline physics-

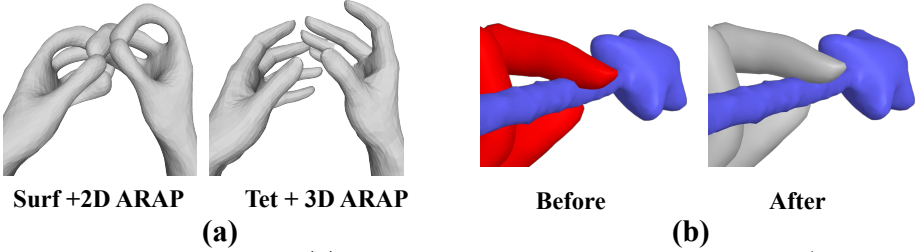


Fig. 7: Ablation studies. (a) **Geometry:** Our tetrahedral formulation (Tet + 3D ARAP) prevents the severe geometric flattening and volume loss seen in surface-only baselines. (b) **Collision:** Our offline collision optimization successfully resolves non-physical inter-penetrations.

based collision penalty is essential to resolve non-physical inter-penetrations, yielding plausible contact mechanics without sacrificing rendering quality.

Appearance Decoupling. As shown in Fig. 3, our physics-based model successfully factorizes the rendering into diffuse and specular components. Explicitly modeling view-dependent highlights prevents lighting artifacts from erroneously baking into the geometry. While standard 3DGS frequently warps surfaces or creates unnatural bumps to fake specular glints, our strict disentanglement eliminates these structural hallucinations. This yields pristine, noise-free normal maps, providing the geometric and photometric purity essential for extracting high-quality, relightable 4D digital assets.

Number of Camera Views. We also evaluate the system’s performance under varying numbers of input camera views. In this ablation, the view restriction is strictly applied end-to-end. As shown in Fig. 6(b), reconstruction quality scales predictably with the camera count. Notably, even in a sparse 4-view setup, the strong spatiotemporal priors from HOST prevent the optimization from collapsing, maintaining a PSNR of nearly 27.89 dB. This confirms that our framework remains highly robust under severe view scarcity.

5 Conclusion

We present a unified framework that bridges accurate kinematic tracking and photorealistic novel view synthesis for 4D hand-object interactions. By coupling a Hand Object Spatiotemporal Transformer (HOST) with a Physics-aware Gaussian (HOPG) module, our method robustly resolves the severe geometric flattening and volume loss common in unconstrained rendering. Furthermore, explicitly decoupling diffuse and specular reflections yields clean, noise-free normal maps, protecting the underlying 3D assets from baked-in lighting artifacts. Currently, our pipeline assumes rigid objects. Additionally, because the HOST prior is trained exclusively on data from our custom capture, it is tightly coupled to our specific camera distributions and backgrounds, limiting immediate generalization to arbitrary public datasets. Future work will integrate non-rigid deformation fields for highly articulated items and diversify the training corpus to scale HOST toward “in-the-wild” videos, advancing toward a universal engine for 4D asset creation.

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